

Zingerone induced caspase-dependent apoptosis in MCF-7 cells and prevents 7,12-dimethylbenz(a)anthracene-induced mammary carcinogenesis in experimental rats

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Abstract

Breast cancer is a prevalent of tumoregenesis in women and reports for the maximum mortality and morbidity in the global. Ginger (*Zingiber officinale*) is the mainly widespread spice and herbal remedies used in the world. Since antique periods, ginger has been used in Greece, India and China for the curing of upset stomach, nausea, diarrhea, colds, and headaches. The current work was planned to explore the anticancer properties of zingerone (ZO) toward 7,12-dimethylbenz(a)anthracene (DMBA)-treated mammary carcinogenesis in Sprague-Dawley (SD) rats and MCF-7 mammary cancer cells. The mammary carcinogenesis was produced through a single dosage of DMBA (20 mg/kg bwt) mixed in soya oil (1 mL) administrated intragastrically with a gavage. We found improved concentrations of lipid peroxidation (LOOH and TBARS), carcinoembryonic antigen, lowered levels of enzymatic (CAT, GPx, and SOD), and nonenzymatic (vitamin E, GSH, and vitamin C) antioxidant in mammary tissues and plasma of DMBA-induced cancer bearing animals. Moreover, augmented concentrations of phase I (Cyt-b₅ and CYP₄₅₀) and reduced levels of phase II (GR and GST) detoxification microsomal proteins in mammary tissues were noticed. ZO administrations significantly reverted back to all these parameters in this way, showing efficient of anticancer effect. Furthermore, our in vitro study also supported the anticancer effect of the treatment of ZO were noticed loss of cell viability, improved reactive oxygen species formation, and reduced MMP. Furthermore, the status of apoptosis proteins such as Bcl-2, Bax, and Bid expressions was determined by using Western blot analysis techniques. Overall, these results proposed the anticancer effect of ZO toward DMBA-induced mammary cancer in SD animals and Michigan cancer foundation-7 mammary cancer cells.

KEYWORDS

antioxidant, anticancer, reactive oxygen species, zingerone, MCF-7 cell lines, 7,12-dimethylbenz(a)anthracene

1 | INTRODUCTION

Polycyclic aromatic hydrocarbons (PAHs) are a product which produced by partial burning of organic material. The principal resources of PAHs contain domestic and industrial oil furnaces, diesel engines and gasoline. The PAHs are extensively disseminated in our surroundings and are initiated in different varieties of cancer. The enzymatic stimulation of PAHs is stimulating to the production of reactive oxygen species (ROS) such as superoxide anion radicals and peroxides, which stimulate oxidative stress in the appearance of lipid peroxidation.^[1] Toxic expression of 7,12-dimethylbenz(a)anthracene (DMBA) is related with its oxidative metabolism more important to the production of reactive metabolites (quinines and epoxides) competent of producing free radicals. The biotransformation enzymes are involved not only in the metabolism of naturally occurring substances, but also contribute to the metabolism of different synthetic substances and chemicals.^[2] Xenobiotics are a general name characterize various unknown chemicals which containing food and drugs that usually exist outside the living being, but to which a living being could be exposed and it may apply their pathological property during the production of ROS, which is associated with the etiology of cancer. The metabolic process of xenobiotics is present with the phase I and phase II detoxification enzymes, which can make a significant role by transferring of biologically inactive substance into toxic or active metabolites and finally removed from the body.^[3]

Breast cancer is a prevalent of tumorigenesis in women and reports for the maximum mortality and morbidity in the global.^[4] The breast cancer etiology is undoubtedly multifactorial and various risk factors containing post menopausal obesity, late menopause, early menarche, family history, hormone replacement therapy, extensive use of oral contraceptives, late age at first birth, nulliparity, and previous benign breast disease.^[5] Oxidative stress stimulated during the production of free radicals or/and decreased the level of antioxidant in the tissues and cancer cells have been playing a main role in the formation tumorigenesis. Enlarged occurrence of oxidative stress and lipid peroxidative markers are initiated in the carcinogenic progression.^[6] In nature, there is an energetic balance between the level of free radicals produced in the body and protection of antioxidant mechanism that scavenge or quench them and in this manner defend the body against pathogenesis and tumor formations.^[7]

Researchers have suggested that different bioactive phytochemicals and nutritional agents exhibit cytotoxic effects against mammary tumor cells in vitro and inhibit the incidence of mammary tumors or/and inhibit the development of tumors in vivo.^[8] Ginger (*Zingiber officinale*) is the mainly widespread spice and herbal remedies used in the world. Since antique periods, ginger has been used in Greece, India, and China for the curing of upset stomach, colds, diarrhea, nausea, and headaches.^[9] Ginger has an energetic phenolic group of compounds such as shogaols, paradols, gingerols, and zingerone. Previous research works demonstrated that these compounds have antiangiogenic, antioxidant, anticancer, and

antiatherosclerotic activities.^[10] Zingerone (ZO) is principally present in dry ginger, as gingerol is changed to zingerone during drying or cooking and it has been reported that the anticancer potential previously.^[11-13] However, phenolic compounds are determined to apply chemopreventive property, but the anticancer potentials of ZO are nevertheless to be comprehensively studied. The target of our present assessment is the anticancer activity of ZO on mammary cancer treated by DMBA in Sprague-Dawley (SD) rats and MCF-7 mammary cancer cell lines.

2 | MATERIALS AND METHODS

2.1 | Chemicals and reagents

Zingerone and DMBA were procured from Sigma Chemical Co., St. Louis. Dulbecco's modified Eagle's medium (DMEM), dimethyl sulfoxide (DMSO), ethidium bromide, fetal bovine serum, and 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide (MTT) were obtained from Himedia Lab Ltd., Mumbai, India. The primary antibodies for Bcl-2, Bid, β -actin, and Bax were acquired from Santa Cruz Biotechnology.

2.2 | Induction of carcinogenesis

The mammary cancer was produced by a single dose of DMBA (20 mg/kg bwt) with mixed in soya oil (1 mL) administrated intragastrically with a gavage as described previously.^[14]

2.3 | Animal model

The adult female rats of SD strain about 140 to 180 g were procured. They were kept up under the normal circumstances of temperature ($25 \pm 2^\circ\text{C}$), humidity and light (12 hour light/dark). The current study was approved by the IAEC, controlled by the CPCSEA. They were fed with regular animal pellet and water.

2.4 | Experimental design

The animals were separated into four groups and each containing six rats. The group I rats were served as a normal food and water ad libitum throughout the experimental period (16 weeks). The group II and III rats were induced mammary cancer with DMBA (20 mg/kg bwt) in 1 mL of soya oil. The group III animals were administrated with ZO one time in a day by initial one week before the cancer induction. Group IV animals were administrated ZO alone the whole time of the investigation time.

The initial and final body weight of the entire groups of animals was documented. At the end of the experimental time rats were fasted overnight and killed by cervical decapitation. The blood was gathered in heparinized tubes and the plasma employed for the various biochemical assays. The mammary tissue was homogenized with proper buffer and centrifuged at 3000g. After centrifugation, the supernatant was analyzed for different biochemical assessment.

Mammary tissue was preserved in formalin (10%) and stored (-80°C) for analysis of histological study.

2.5 | Determination of enzyme linked immunosorbent assay method

The tumor marker enzyme of CEA was analyzed by enzyme linked immunosorbent assay (manufacturers of R&D systems) method.

2.6 | Biochemical analysis

The level of mammary tissue and plasma LOOH was determined through the way of Jiang et al.^[15] The level of mammary tissue and plasma thiobarbituric acid reactive substances (TBARS) was determined through the way of Ohkawa et al.^[16] The activity of GPx was analysed through the way of Rotruck et al.^[17] The GST was analyzed through the way of Habig et al.^[18] The activities of CAT and SOD were determined by the kit method. The level of GSH was quantified by the way of Ellman.^[19] The GR was analysed by the way of Carlberg and Mannervik.^[20] Cyt-b₅ and CYP₄₅₀ microsomal protein levels in mammary tissues were analysed through the way of Omura and Sato.^[21] The vitamin E level was analysed by the way of Desai.^[22] The vitamin C level was determined through the way of Omaye et al.^[23]

2.7 | Histopathological examination

The portion of mammary tissue samples in each group was located in neutral buffered formalin (10%), then sectioned, stained with hematoxylin and eosin, and observed by the microscope.

2.8 | Cell culture

Michigan cancer foundation-7 (MCF-7) a human breast cancer cell lines were procured. Cells were cultured in DMEM and kept in a humidify environment (37°C), including air incubation (95%) and CO_2 (5%). ZO was prepared by dissolving with 0.5% DMSO. An equivalent quantity of DMSO was used as a normal control.

2.9 | Cell viability assay

The MTT assay was used to determine the toxic effect of ZO toward cultured cells.^[24] The cells were seeded in 96-well plates and permitted to grow confluence of 70 to 80%. The MCF-7 cells were added with various concentrations (5, 10, 15, 20, 25, 30, 35, 40, 45, and $50\text{ }\mu\text{M}$) of ZO for 24 hour with DMSO. After removal the medium, $100\text{ }\mu\text{L}$ of MTT was added and kept at 37°C for 1 hour. The absorbance was identified by a microplate reader at 550 nm with a 650 nm.

2.10 | Measurement ROS generation

The intracellular ROS production was measured in MCF-7 cells using the dichloro-dihydro-fluorescein diacetate (DCFH-DA) staining

method.^[25] The DCFH-DA is a cell permeable non fluorescent probe, which has been applied as a substrate for an amount of intracellular oxidant formation. After treating with various dosages (5, 10, and $15\text{ }\mu\text{M}$) of ZO, cells were stained with DCFH-DA ($10\text{ }\mu\text{M}$) for 30 min (37°C). The cells were considered at an emission/excitation wavelength of 485/535 nm in a microplate reader.

2.11 | Measurement of MMP

The MMP was evaluated by Rh-123, lipophilic cationic dye.^[26] The cells (1×10^6 cells/well) were added in six wells plate and treated with various dosages (5, 10, and $15\text{ }\mu\text{M}$) of ZO. After 24 hrs treatment, the cells were kept by way of Rh-123 dye for 30 minute. The MMP was estimated under a fluid cell imaging station at 485/530 nm under spectrofluorometer.

2.12 | Western blot analysis

The MCF-7 cells were added by ZO at various dosages (5, 10, and $15\text{ }\mu\text{M}$) and lysed with appropriate lysis buffer. The proteins were quantified and separated by using SDS-PAGE gel electrophoresis. Afterward, proteins were transferred onto a polyvinyl difluoride membrane, blocked with blocking buffer (5% bovine serum albumin) and kept with particular primary antibodies for Bcl-2 (mouse monoclonal; 1:1000), Bid (mouse monoclonal; 1:1000) and Bax (mouse monoclonal; 1:1000) at 4°C overnight. The membranes were washed with TBS and kept with suitable secondary antibodies.

2.13 | Statistical analysis

The data are characterized as the mean $\pm$ standard deviation. The variations between the groups were observed using the one-way analysis of variance test. The $P < .05$ was measured statistically significant.

3 | RESULTS

3.1 | Effect of ZO on the changes of body weight

Table 1 revealed the changes of body weight in experimental and control animals. Initially, no significant body weight modification was determined in experimental and control rats. The body weight was considerable ($P < .05$) decreased as an ending of the experiment of DMBA-induced rats when compared with the normal rats. Conversely, ZO treatment prevented the considerable ($P < .05$) decreased in the body weight of DMBA-treated rats. ZO alone and control rats showed no body weight alterations were observed.

3.2 | Effect of ZO on lipid peroxidative markers and antioxidant activities

Tables 2,3 demonstrates the effect of ZO on lipid peroxidation (TBARS and LOOH), antioxidant (GPx, vitamin C, SOD, GSH, CAT,

TABLE 1 Initial and final body weight changes of control and experimental animals in each group

Treatments	Initial body weight (g)	Final body weight (g)
Control	143.50 ± 9.36	178.86 ± 12.30
DMBA	146.26 ± 8.35	125.91 ± 9.28*
DMBA + ZO (20 mg/kg bwt)	148.73 ± 9.58	176.54 ± 14.40 [#]
ZO alone (20 mg/kg bwt)	144.18 ± 8.89	177.48 ± 11.85

Note: Statistical significance was compared within the group as follows:

*Control rats; [#]DMBA induced rats. Data are expressed as mean ± standard deviation for n = 6 animals in each group. Values not sharing a common superscript differ significantly at $P < .05$.

Abbreviations: DMBA, 7,12-dimethylbenz(a)anthracene; ZO, zingerone.

and vitamin E) in plasma and mammary tissues of experimental and control rats, respectively. DMBA-induced rats showed notably ($P < .05$) improved lipid peroxidation and reduced antioxidant activities when compared to normal animals. Whereas, treatment with ZO notably ($P < .05$) reduced lipid peroxidation and improved antioxidants as compared with tumor bearing animals. There were no considerable ($P < .05$) changes were documented in normal and drug alone treated rats.

TABLE 2 The levels of lipid peroxidation and antioxidants in plasma of control and experimental animals in each group

Treatments	TBARS, nmol/mL	LOOH, nmol/mL	SOD (U*/mL)	CAT (U [#] /mL)	GPx (U ^{\$} /mL)	GSH (mg/dL)	vitamin E (mg/dL)	vitamin C (mg/dL)
Control	3.69 ± 0.21	1.51 ± 0.11	6.06 ± 0.46	3.87 ± 0.28	88.23 ± 6.69	41.20 ± 3.75	2.71 ± 0.19	2.62 ± 0.19
DMBA	7.85 ± 0.64*	3.17 ± 0.23*	2.90 ± 0.18*	2.01 ± 0.17*	59.50 ± 2.47*	19.93 ± 1.34*	1.13 ± 0.09*	1.23 ± 0.08*
DMBA + ZO (20 mg/kg bwt)	4.06 ± 0.32 [#]	1.79 ± 0.13 [#]	5.49 ± 0.32 [#]	3.31 ± 0.24 [#]	83.16 ± 7.48 [#]	40.71 ± 3.34 [#]	2.52 ± 0.17 [#]	2.36 ± 0.16 [#]
ZO alone (20 mg/kg bwt)	3.90 ± 0.33	1.66 ± 0.14	6.01 ± 0.49	3.51 ± 0.19	86.76 ± 6.36	41.00 ± 2.95	2.68 ± 0.21	2.48 ± 0.21

Note: Statistical significance was compared within the group as follows: *Control rats; [#]DMBA-induced rats. Data are expressed as mean ± standard deviation values for six animals in each group. Values not sharing a common superscript letter differ significantly at $P < .05$. Units of SOD*, CAT[#], GPx^{\$} are amount of enzyme required for inhibition of 50% NBT reduction, micromoles of H₂O₂ utilized/second, and micromoles of glutathione utilized/minute, respectively.

Abbreviations: DMBA, 7,12-dimethylbenz(a)anthracene; TBARS, thiobarbituric acid reactive substances; ZO, zingerone.

TABLE 3 The levels of lipid peroxidation and antioxidants in mammary tissues of control and experimental animals in each group

Treatments	TBARS (nmol/100 mg protein)	LOOH (nmol/100 mg protein)	SOD (U*/mg protein)	CAT (U [#] /mg protein)	GPx (U ^{\$} /mg protein)	GSH (mg/100 mg tissue)	Vitamin E (mg/100 mg protein)	Vitamin C (mg/100 mg protein)
Control	2.32 ± 0.18	1.40 ± 0.09	17.36 ± 1.26	48.52 ± 3.15	15.68 ± 1.21	19.34 ± 1.58	4.02 ± 0.31	3.79 ± 0.31
DMBA	5.71 ± 0.42*	3.76 ± 0.23*	8.67 ± 0.67*	26.79 ± 1.92*	8.38 ± 0.62*	8.86 ± 0.62*	2.37 ± 0.19*	1.54 ± 0.12*
DMBA + ZO (20 mg/kg bwt)	2.70 ± 0.23 [#]	1.68 ± 0.12 [#]	16.12 ± 1.54 [#]	44.72 ± 3.74 [#]	15.08 ± 1.38 [#]	18.80 ± 1.31 [#]	3.65 ± 0.31 [#]	3.58 ± 0.32 [#]
ZO alone (20 mg/kg bwt)	2.54 ± 0.21	1.44 ± 0.14	16.90 ± 1.19	46.56 ± 3.43	15.23 ± 1.06	19.16 ± 1.35	3.96 ± 0.29	3.67 ± 0.25

Note: Statistical significance was compared within the group as follows: *Control rats; [#]DMBA induced rats. Data are expressed as mean ± standard deviation values for six animals in each group. Values not sharing a common superscript letter differ significantly at $P < .05$. Units of SOD*, CAT[#], GPx^{\$} are amount of enzyme required for inhibition of 50% NBT reduction, micromoles of H₂O₂ utilized/second and micromoles of glutathione utilized/minute, respectively.

Abbreviations: DMBA, 7,12-dimethylbenz(a)anthracene; TBARS, thiobarbituric acid reactive substances; ZO, zingerone.

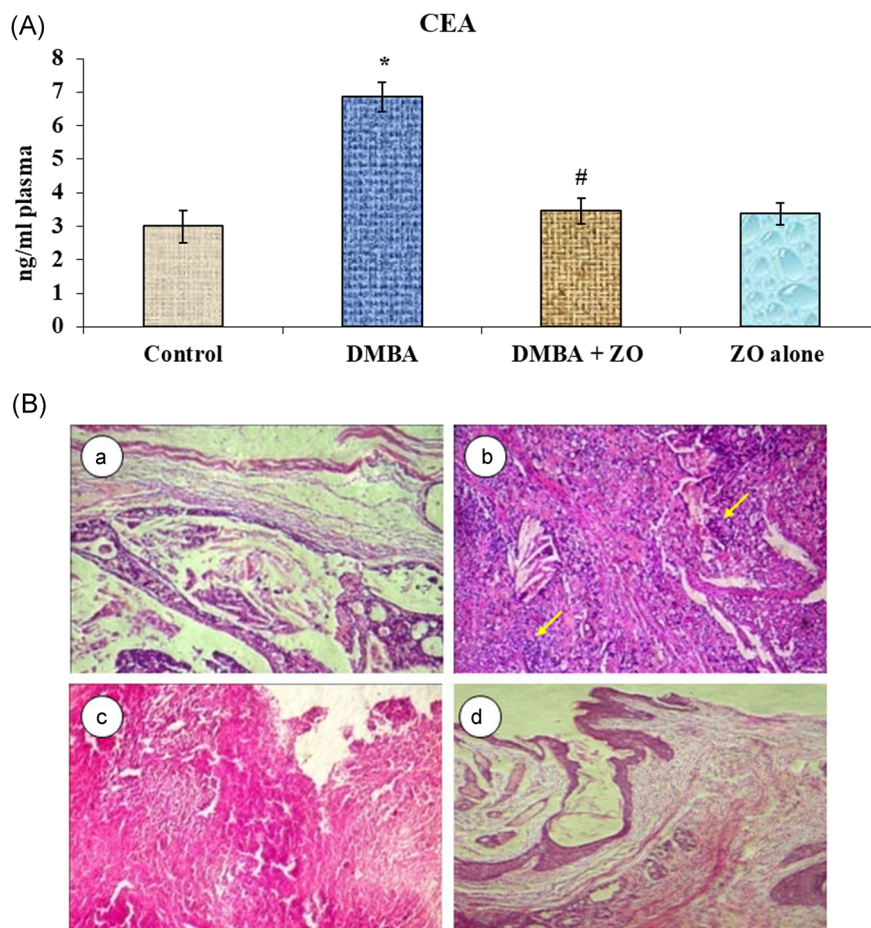
3.3 | Effect of ZO on tumor marker enzyme

Figure 1A shows the level tumor marker enzyme (CEA) in experimental and control animals. DMBA-treated rats showed notably ($P < .05$) augmented CEA level when compared to normal animals. Conversely ZO treatments considerable ($P < .05$) reduced levels of CEA when compared with cancer bearing rats. No considerable ($P < .05$) alterations were observed in the control and compound alone treated rats.

3.4 | Effect of ZO on biotransformation enzyme levels

Tables 4 show the activities of phase I (CYP₄₅₀ and Cyt-b₅) and phase II (GST and GR) detoxification enzymes in mammary tissue of experimental and control rats. The levels of phase I enzymes were notably ($P < .05$) augmented, while the activities of phase II enzymes were considerable ($P < .05$) lowered in DMBA-induced rats when difference with the control animals. Oral supplementation of ZO were notably ($P < .05$) reduced in phase I enzymes and considerably ($P < .05$) improved in phase II enzymes when difference with cancer bearing rats. Conversely, there were no considerable ($P < .05$) changes in the rats supplemented with ZO alone when compared to normal rats.

FIGURE 1 A, Effect of ZO on CEA level of control and experimental mice. Statistical significance was compared within the group as follows: *Control rats; #DMBA induced rats. All the values are expressed as mean $\pm$ SD ($n = 6$). Values not sharing a common superscript differ significantly at $P < .05$. B, The histopathological analysis of mammary tissues in control and experimental animals using H&E staining (a) Vehicle control animals showing normal ductal epithelium. (b) DMBA-induced mammary cancer bearing animals showed invasive ductal adenocarcinoma. (c) ZO-treated cancer bearing animals showed near normal architecture as that of vehicle control animals. (d) ZO drug control animals showing normal architecture. Arrows indicate adenocarcinoma. CEA, carcinoembryonic antigen; DMBA, 7,12-dimethylbenz(a)anthracene; H&E, hematoxylin and eosin; SD, standard deviation; ZO, zingerone



3.5 | Histopathological alteration of mammary tissue

Figure 1B reveals the histopathological assessment of mammary tissues in control and DMBA induced rats. DMBA induced (B) cancer bearing animal's demonstrated changed ductal epithelial lining representing invasive ductal adenocarcinoma. ZO(C) treated tumor bearing animals explained deterioration in tumor with better ductal architecture. The control (A) and ZO(D) only administrated rat's demonstrated regular ductal epithelial architecture.

3.6 | Effect of ZO on MTT analysis

Figure 2 shows the cell viability was analyzed on MCF-7 cells with the various concentrations (5, 10, 15, 20, 25, 30, 35, 40, 45, and

50 μ M) of ZO for 24 hour by the MTT assay. The IC_{50} value of ZO at a concentration of $15.53 \pm 1.26 \mu$ M for MCF-7 cells after 24 hour treatment was analysed from the growth inhibition curve and we have chosen for 5, 10, and 15 μ M concentrations of ZO for the further analysis.

3.7 | Effect of ZO on ROS staining

Figure 3A-B shows the production ROS was analysed by DCFH-DA staining method. Cells treated with different concentrations of ZO (5, 10 and 15 μ M) demonstrates improved the levels of ROS production which indicates huge green fluorescence intensity (b-d), while normal control cells (a) shows small green fluorescence were observed.

TABLE 4 The levels of biotransformation enzymes in mammary tissues of control and experimental animals in each group

Treatments	CYP ₄₅₀	Cyt-b ₅	GST	GR
Control	1.70 $\pm$ 0.12	1.59 $\pm$ 0.09	2.66 $\pm$ 0.18	5.38 $\pm$ 0.38
DMBA	3.86 $\pm$ 0.24*	3.98 $\pm$ 0.27*	1.07 $\pm$ 0.09*	2.75 $\pm$ 0.18*
DMBA + ZO (20 mg/ kg bwt)	2.12 $\pm$ 0.19#	2.05 $\pm$ 0.15#	2.41 $\pm$ 0.18#	5.26 $\pm$ 0.39#
ZO alone (20 mg/ kg bwt)	1.86 $\pm$ 0.15	1.72 $\pm$ 0.11	2.56 $\pm$ 0.21	5.32 $\pm$ 0.42

Statistical significance was compared within the group as follows: *Control rats; #DMBA induced rats. Data are expressed as mean $\pm$ standard deviation values for six animals in each group. Values not sharing a common superscript letter differ significantly at $P < .05$. CYP₄₅₀-nmol/mg microsomal protein. Cyt-b₅-nmol/mg microsomal protein. GST- μ mol of CDNB-GSH conjugate formed/mg microsomal protein/min. GR- μ mol of NADPH oxidized/mg microsomal protein/min.

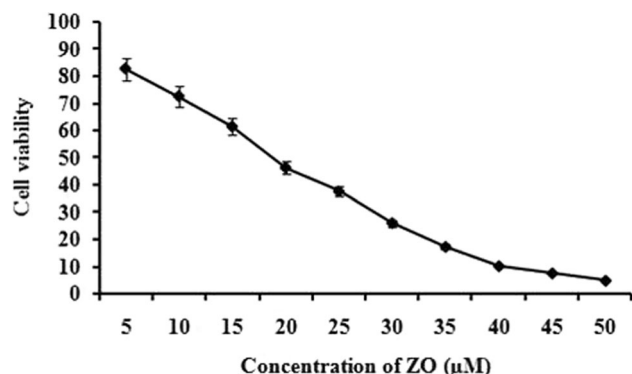


FIGURE 2 The cytotoxicity effect of ZO on MCF-7 cells was measured by the MTT assay. MCF-7, Michigan cancer foundation-7; MTT, 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide; ZO, zingerone

3.8 | Effect of ZO on MMP staining

Figure 4A-B shows the alteration of MMP was analysed by Rh-123 lipophilic cationic dye. The polarized mitochondria membrane emit elevated density of green fluorescence was noticed in untreated MCF-7 cells (a). Conversely, MCF-7 cells treated with different dosages (5, 10 and 15 μM) of ZO for 24 hrs demonstrate reduced green fluorescence intensity (b-d).

3.9 | Effect of ZO on the western blot analysis protein expression of apoptotic markers

Figure 5 illustrates the apoptotic protein expression of control and ZO-treated MCF-7 cells. The control cells showed improved expression of Bcl-2, whereas lowered expression of Bax and Bid

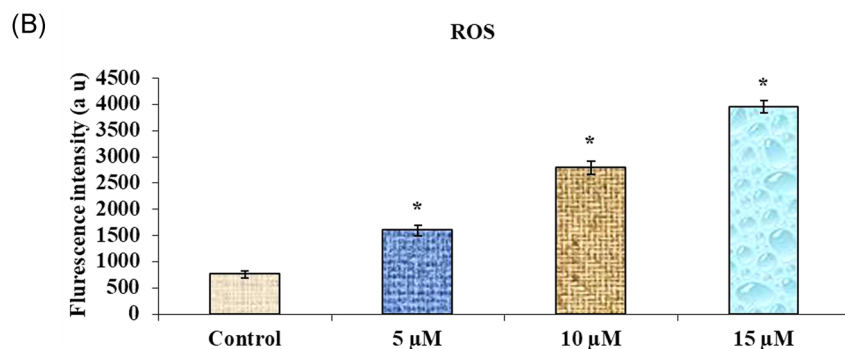
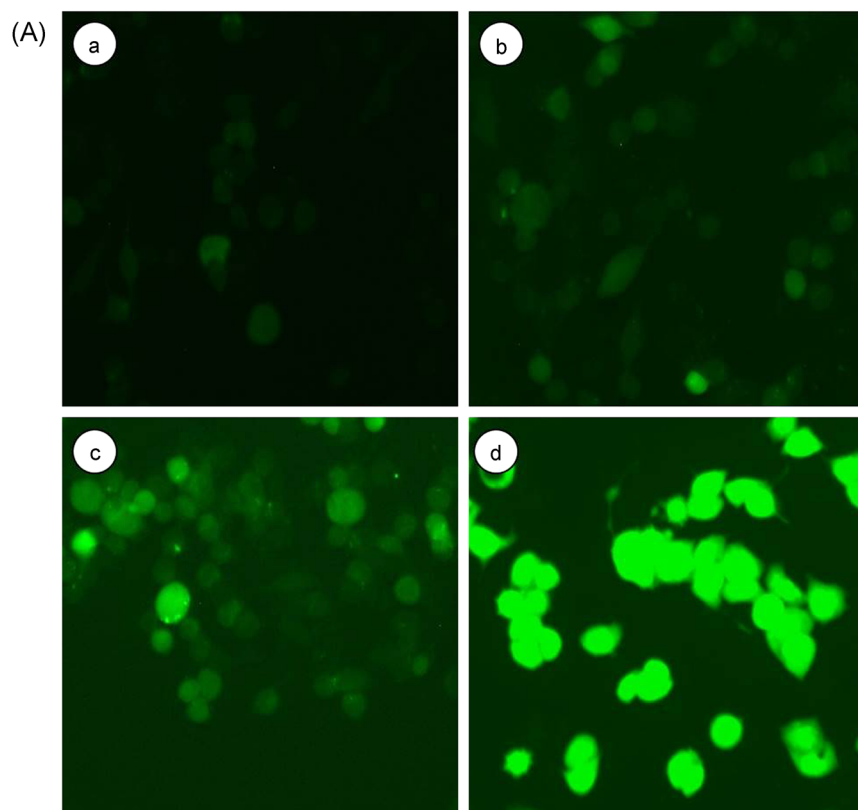
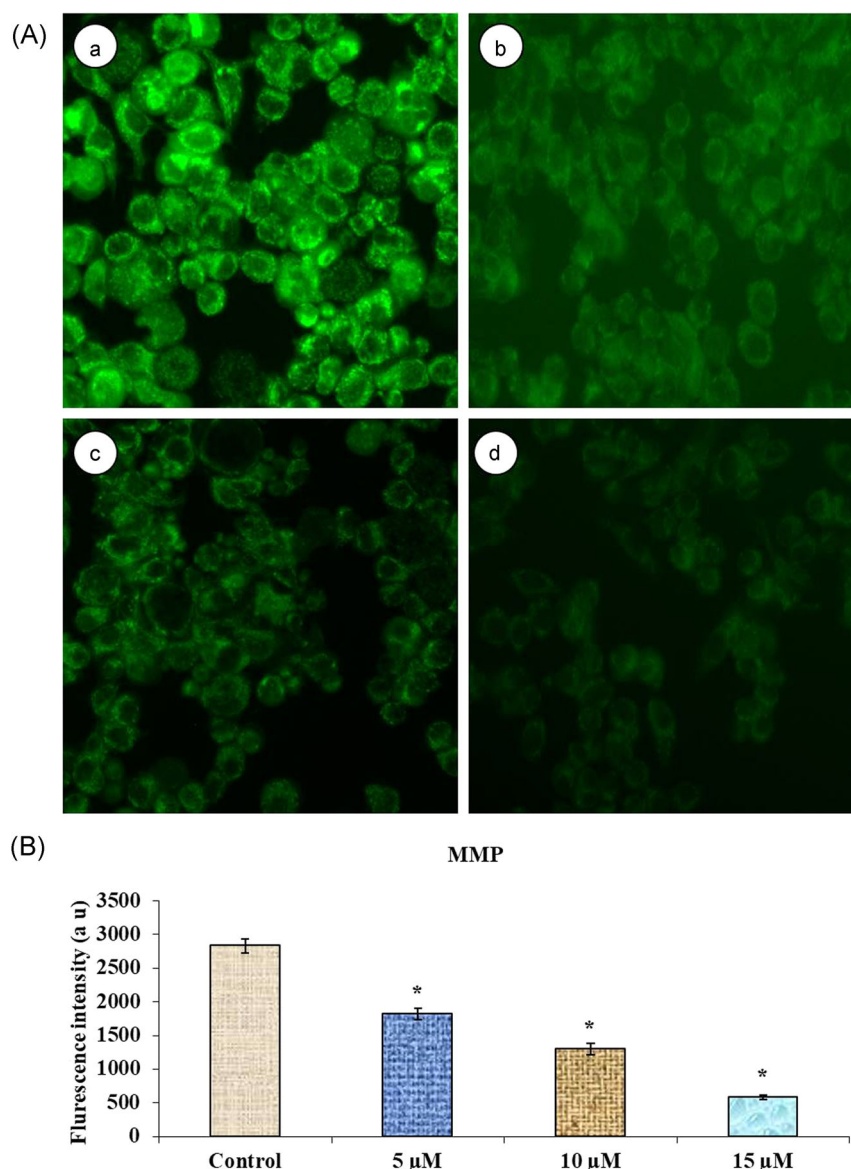


FIGURE 3 The effect of ZO on intracellular ROS generation was evaluated with MCF-7 cells by using DCFH-DA staining. A, Depicted fluorescence intensity was detected by spectrofluorometer. B, Data represent mean $\pm$ SD of triplicate, * $P < .05$ as compared with the control group. DCFH-DA, dichloro-dihydro-fluorescein diacetate; MCF-7, Michigan cancer foundation-7; ROS, reactive oxygen species; SD, standard deviation

FIGURE 4 The effect of ZO on MMP damage was evaluated with MCF-7 cells using the Rh-123. A, Depicted fluorescence intensity was detected by spectrofluorometer. B, Data represent mean $\pm$ SD of triplicate, $*P < .05$ as compared with the control group. MCF-7, Michigan cancer foundation-7; MMP, matrix metalloproteinase; SD, standard deviation; ZO, zingerone



were found. Conversely, ZO (5, 10, and 15 μ M) treated cells revealed downregulated the antiapoptotic protein and improved the expression of proapoptotic proteins.

4 | DISCUSSION

The total body weight of cancer bearing rats was reduced due to the alterations in energy metabolism.^[27] Furthermore, ZO-treated animals gradually improved body weight, which evidence showed that the reduced level of lipid peroxidation in ZO treated rats could be accredited to the free radical scavenging activity. The histopathological assessment of mammary tissues in DMBA induced animal's demonstrated changed ductal epithelial lining representing invasive ductal adenocarcinoma. ZO treated tumor induced rats explained deterioration in tumor with better ductal architecture, demonstrating the anticancer properties of ZO. The normal and ZO only administrated rat's were demonstrated regular ductal epithelial

architecture indicative of the non-toxic effect of ZO. Previous study also reported that the ZO has been inhibiting the decreasing of the body weight in a mouse tumor model.^[12]

Free radicals are frequently formed in rats induced by carcinogens. These free radicals arbitrated oxidative stress is playing a main function in all stages of cancer.^[28] The accurate balance between antioxidants and LPO should have been maintained in the cells because of their prospective significance in the pathogenesis of different pathologic diseases such as cancer.^[29] An inequality between the antioxidants and ROS occurs oxidative stress due to the formation of reactive oxygen is larger than the cells' capability to detoxify the reactive intermediates, which show the way to oxidative injury to molecules.^[30] The production of antioxidants is a most important mechanism for defending cells against a diversity of exogenous and endogenous toxic chemicals of DMBA and ROS.^[31] In the current work, the decreased activities of these antioxidant enzymes were found in cancer induced rats and this could be owing to these enzymes are not enough to scavenging

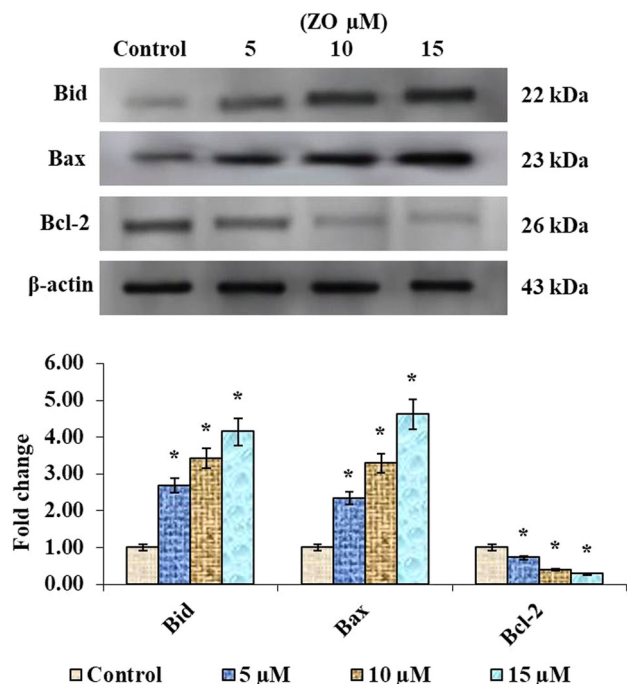


FIGURE 5 The anticancer effect of ZO of apoptotic signaling proteins in MCF-7 cell lines was examined by western blot analysis technique. The cells were treated with ZO (5, 10, and 15 μ M) for 24 hours and dose-dependent changes in the expressions of Bcl-2, Bax, and Bid were analysed. β -Actin was used as a loading control. Data represent mean $\pm$ SD of triplicate, * P < .05 as compared with the control group. MCF-7, Michigan cancer foundation-7; SD, standard deviation; ZO, zingerone

of ROS.^[32] Treated with ZO for cancer bearing animals showed the oxidant/antioxidant activities were reverted back to the nearest standard levels. This could be due to the powerful antioxidant action of ZO and this is well in concurrence with the earlier report in hepatic toxicity.^[33]

Free radicals participate a vital function in tumor initiation by modification of cellular metabolic processes or direct chemical reaction. The stimulation of carcinogen is mainly catalyzed by phase I (CYP₄₅₀ and Cyt-b₅) enzymes^[34] and detoxified by the presence of phase II (GST and GR) enzymes thereby protecting against carcinogenic injury.^[35] The scavengers of phase II enzymes represent of inhibitors at all stages of carcinogenesis. These enzymes are present in cytosolic and mitochondrial roles primarily implicated in the detoxification and biotransformation of carcinogens.^[36] The level of phase I enzymes were remarkably augmented and phase II enzymes reduced in cancer bearing rats were observed in the current study is good concordance with this earlier report.^[37] Cancer bearing animals was treated with ZO those enzyme activities were altered near to normal levels during the suppression of phase I and stimulation of phase II enzymes by ZO. The CEA is a tumor associated glycoprotein that has been generally considered its prospective function as a prognostic indicator.^[38] A high level of the CEA in cancer induced rats may relate to the initiation and progression of tumor.^[39] In the current

study an improved level of CEA was found in cancer induced rats and the same results were informed previously.^[40] ZO treatment with cancer induced animals reverted back to the CEA level due to the anti-tumor activity.

The MTT assay is the most commonly used methods for the detection of cell viability and/or cytotoxicity for anticancer drugs.^[41] The hopeful anticancer mediator should have convinced cytotoxic characteristic. It means a drug is only toxic to cancer cells lacking any difficulties to normal cells. In this present study, we attempt to estimate the specificity of ZO as anticancer agents. Earlier reports have revealed the efficacy of ZO against neuroblastoma cells.^[13] Mitochondria play a main function in the activation of apoptosis, ZO is well-known to stimulate ROS and consequently leading to modification in MMP. The permeability transition through oxidative stress in MMP and an opening of the mitochondrial outer membrane channel permit cytochrome c release into the cytosol, which is normally activated cascades of caspases and indirectly or directly for activating apoptosis.^[42] The current study has revealed the alteration of MMP in ZO-induced MOF tosis in cancer cells. Earlier study also reported that the Ginger leaf induced apoptosis in HCT116 cell lines were supported by the present study.^[43]

5 | CONCLUSION

From our current research, the applications of ZO provide an apparent motivation about a suppress result of mammary cancer and it also had protecting consequences of chemically induced mammary cancer during the inhibition of lipid peroxidation, altering phase II and phase I detoxification enzymes and rising the activities of nonenzymatic and enzymatic antioxidants in vivo. Additionally, it also supported by the histological analysis. Moreover, this study evidently shows the anticancer potential of ZO in decreasing MMP and stimulating ROS production during the activation of apoptosis by in vitro study. Overall, these results confirm the anticancer prospective of ZO against DMBA-treated mammary cancer in SD rats and MCF-7 mammary cancer cell lines.

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CONFLICT OF INTERESTS

The authors declare that there are no conflict of interests.

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